

Expander-Sparsified Tensor Embeddings Give a Deterministic Rank-One Matrix RIP

Discovery packet for arXiv:2310.18565
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Status. Candidate substantial partial result, likely valid, requiring expert review. It answers the source challenge for matrices of rank one, over both $\mathbb{R}$ and $\mathbb{C}$, with rank-one measurement matrices and $m = n^{1+o(1)}$. The case of rank $r > 1$ remains open.

1 Source and exact challenge

The source is Simon Foucart, *Linearly Embedding Sparse Vectors from ℓ_2 to ℓ_1 via Deterministic Dimension-Reducing Maps*, arXiv:2310.18565, Section 6, page 14 [1]. It asks for a deterministic linear map

$$\mathcal{A} : \mathbb{K}^{n \times n} \longrightarrow \mathbb{K}^m$$

such that, for all matrices X of rank at most r ,

$$(1 - \delta)\|X\|_F \leq \|\mathcal{A}(X)\|_1 \leq (1 + \delta)\|X\|_F, \tag{1}$$

in a regime $m \asymp n^\lambda r^\mu \ll n^2$, preferably with rank-one measurement matrices. The source calls this a “wide open question.”

gaussian entries, see [23]. But, as in the vector case, no deterministic linear map $\mathcal{A} : \mathbb{K}^{n \times n} \rightarrow \mathbb{K}^m$ is known in this optimal regime $m \asymp nr$. Worst, I am not aware of simple deterministic constructions overcoming the trivial regime $m \asymp n^2$ and I have not seen an effective analog of the notion of coherence.

Interestingly, the theory of low-rank recovery can also be built from a modification of the RIP featuring the ℓ_1 -norm as the inner norm, see [5 Chapter 16], namely from

$$(1 - \delta)\|X\|_F \leq \|\mathcal{A}(X)\|_1 \leq (1 + \delta)\|X\|_F \quad \text{for all rank-}r \text{ } X \in \mathbb{K}^{n \times n}. \quad (19)$$

This alternative version seems even more relevant in the present scenario. Indeed, when $m \asymp nr$, the rank-one measurements

$$\mathcal{A}(Z)_i = \langle b_i, Za_i \rangle = \langle A_i, Z \rangle_F, \quad A_i := b_i a_i^*,$$

with independent properly normalized gaussian vectors $a_1, \dots, a_m, b_1, \dots, b_m \in \mathbb{K}^n$ do not lead to the RIP [18] but to the RIP [19], while the latter still enables low-rank recovery via nuclear norm minimization, see [24]. It also enables low-rank recovery via iterative-thresholding-type algorithms, see [25]. But the possibility of fulfilling this modified RIP in a regime $m \asymp n^4 r^\mu \ll n^2$ with deterministic matrices $A_1, \dots, A_m \in \mathbb{K}^{n \times n}$ —of rank one or even unrestricted—is a wide open question. This question is hereby set as a challenge to the readers.

7 Addendum: isometric embeddings, spherical designs, tensors

As mentioned in Remark 2 isometric embedding from $\ell_2^N(\mathbb{K})$ to $\ell_{2k}(\mathbb{K})$, $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, have connections with spherical designs and tensors. This is made precise by the following result, found in [26] for the case $\mathbb{K} = \mathbb{R}$.

Theorem 5. *Let $2k \geq 2$ be an even integer. The following properties are equivalent:*

- 1) *There exists $A \in \mathbb{K}^{N \times n}$ providing an isometric embedding from ℓ_2^n into ℓ_{2k}^N , i.e.,*

$$\|Ax\|_{2k} = \|x\|_2 \quad \text{for all } x \in \mathbb{K}^n;$$

- 2) *There exist $x_1, \dots, x_N \in S_2^n$ and $\tau_1, \dots, \tau_N \geq 0$ with $\tau_1 + \dots + \tau_N = 1$ such that*

$$\sum_{i,j=1}^N \tau_i \tau_j |\langle x_i, x_j \rangle|^{2k} = \int_{S_2^n \times S_2^n} |\langle x, y \rangle|^{2k} d\sigma(x) d\sigma(y),$$

where σ is the normalized standard measure on the unit sphere S_2^n of ℓ_2^n ;

- 3) *There exist $x_1, \dots, x_N \in S_2^n$ and $\tau_1, \dots, \tau_N \geq 0$ with $\tau_1 + \dots + \tau_N = 1$ such that*

$$\sum_{i=1}^N \tau_i \otimes^k (x_i \otimes \bar{x}_i) = \int_{S_2^n} \otimes^k (x \otimes \bar{x}) d\sigma(x).$$

Figure 1: The exact challenge in arXiv:2310.18565, page 14. The image is rendered from the local source PDF included with this packet.

We use the Frobenius pairing $\langle A, X \rangle_F = \text{tr}(A^* X)$. An $M \times n$ matrix B , whose j th row is b_j^* , will be called (a, b) -flat and tight if

$$B^* B = I_n, \quad a\sqrt{M}\|x\|_2 \leq \|Bx\|_1 \leq b\sqrt{M}\|x\|_2 \quad (x \in \mathbb{K}^n). \quad (2)$$

2 Result

Theorem 1 (Rank-one subcase of the modified matrix RIP). *Let $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ and fix $0 < \delta < 1$. There is a deterministic polynomial-time construction, for every sufficiently large n , of rank-one matrices $A_1, \dots, A_m \in \mathbb{K}^{n \times n}$ such that*

$$(1 - \delta)\|X\|_F \leq \sum_{i=1}^m |\langle A_i, X \rangle_F| \leq (1 + \delta)\|X\|_F \quad (3)$$

for every matrix $X \in \mathbb{K}^{n \times n}$ of rank at most one. Moreover,

$$m \leq n \exp(C_\delta (\log \log(n + 2))^2) = n^{1+o(1)} = o(n^2). \quad (4)$$

The finitely many smaller dimensions can be covered deterministically after enlarging the constant depending on δ .

3 Proof intuition

An explicit almost-Euclidean subspace of ℓ_1^M supplies a matrix B whose vector Bx has almost constant ℓ_1 -to- ℓ_2 ratio. The specific recursive extractor construction of Indyk has an additional feature: each stage repeats coordinates of a concatenation of orthonormal bases with constant multiplicity. Hence it is an exact tight frame. After normalization, B obeys (2) with b/a as close to one as desired and $M = n^{1+o(1)}$.

The full tensor coordinates of a rank-one matrix $X = uv^*$ factor:

$$|\langle b_j b_k^*, uv^* \rangle_F| = |b_j^* u| |b_k^* v|.$$

There is therefore no need to keep all M^2 pairs (j, k) . Retain only the edges of a constant-degree spectral expander on the two coordinate sets. The expander mixing lemma compares the retained sum with the product of the two ℓ_1 norms, and tightness controls the error by $\|u\|_2 \|v\|_2 = \|uv^*\|_F$. Choosing the frame distortion and normalized second eigenvalue small enough gives any prescribed δ . For rank $r > 1$, absolute values destroy this separability; that is exactly where the argument stops.

4 Two ingredients

Lemma 2 (Explicit flat tight frames). *For every $0 < \eta < 1/4$ and $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ there are constants $0 < \rho_0 \leq \rho_1 < \infty$, independent of n and η , and a deterministic polynomial-time construction of $B = B_{n,\eta} \in \mathbb{K}^{M \times n}$ such that*

$$B^* B = I_n, \quad (5)$$

$$(1 - \eta)\rho\sqrt{M}\|x\|_2 \leq \|Bx\|_1 \leq (1 + \eta)\rho\sqrt{M}\|x\|_2, \quad (6)$$

for some $\rho \in [\rho_0, \rho_1]$, and

$$M \leq n \exp(C_\eta (\log \log(n + 2))^2). \quad (7)$$

Proof. We record why the standard explicit embedding can be chosen tight; this is the structural point used later. Indyk's Theorem 1.1 and Corollary 5.2 construct an explicit $\ell_2^n \rightarrow \ell_1^M$ embedding with distortion $1 + O(\eta)$ and $M = n \exp(O_\eta((\log \log n)^2))$ by recursively applying extractor-based block maps [2]. At one recursion level its dictionary D is the vertical concatenation of L orthonormal matrices, so $D^*D = LI$. The extractor graph is left d -regular, and the associated map F merely places every coordinate of Dx in exactly d output positions. Consequently

$$F^*F = LdI. \quad (8)$$

After division by $\sqrt{Ld}$, each recursion level is an exact isometry. Indyk's displayed block estimate becomes

$$(1 - O(\zeta))\sqrt{b}\|x\|_2 \leq \sum_{s=1}^b \|(Fx)_s\|_2 \leq \sqrt{b}\|x\|_2 \quad (9)$$

for the normalized map, where the output is split into b blocks.

Iterate (9) on every block for $O(\log \log n)$ levels, taking ζ a sufficiently small multiple of $\eta/\log \log(n+2)$. The product of the lower factors is at least $1 - \eta/2$, while every level remains an exact isometry by (8). At the terminal fixed block dimension, use a deterministic finite spherical quadrature. Concretely, a sufficiently fine finite set on the sphere can be symmetrized under signed coordinate permutations. Its covariance is then exactly scalar, while its equal-weight averages of the equicontinuous family $y \mapsto |\langle x, y \rangle|$ approximate the rotational average uniformly in x . Repeating points clears rational weights. After scalar normalization, this terminal map is tight and has ℓ_1 distortion at most $1 + \eta/2$. Its dimension depends only on η .

Composing the normalized block maps with the terminal maps gives (5). The number of leaves is M ; multiplying the $\sqrt{b}$ factors in (9) gives exactly $\sqrt{M}$ (up to the fixed terminal factor). This proves (6). The rotational first absolute moment times the square root of the terminal dimension is bounded above and below by universal positive constants, giving ρ_0, ρ_1 . Indyk's dimension calculation gives (7). Padding the input to the next admissible dictionary dimension and restricting the columns preserves tightness and changes only the displayed constant.

The construction above is real. To retain the same distortion over $\mathbb{C}$, apply it in real dimension $2n$ and write its row matrix as $R = [P \ Q]$. For $z = x + iy$, put

$$C = P - iQ, \quad B_{\mathbb{C}} = 2^{-1/2}C.$$

Since $R^T R = I_{2n}$, we have $P^T P = Q^T Q = I_n$ and $P^T Q = Q^T P = 0$, hence $C^*C = 2I_n$ and $B_{\mathbb{C}}^* B_{\mathbb{C}} = I_n$. If U_θ is realification of multiplication by $e^{i\theta}$, then, up to the harmless choice of sign of θ ,

$$(RU_\theta z)_j = \operatorname{Re}(e^{i\theta}(Cz)_j).$$

Integrating the real flat estimate in θ and using

$$\frac{1}{2\pi} \int_0^{2\pi} |\operatorname{Re}(e^{i\theta}w)| d\theta = \frac{2}{\pi}|w|$$

shows that $B_{\mathbb{C}}$ has exactly the same relative $1 + O(\eta)$ distortion; only ρ is multiplied by $\pi/(2\sqrt{2})$. The dimension is $(2n)^{1+o(1)}$, proving the complex statement as well. $\square$

Lemma 3 (All-orders explicit expander mixing). *For every $\theta > 0$ there is a constant $d = d(\theta)$ such that, for every sufficiently large M , one can deterministically construct in time polynomial in M a*

d -regular bipartite graph $G = (L, R, E)$ with $|L| = |R| = M$ whose adjacency matrix has all singular values except the top one at most θd . Thus, for all $f, g \in \mathbb{R}^M$,

$$\left| \sum_{(j,k) \in E} f_j g_k - \frac{d}{M} \left(\sum_j f_j \right) \left(\sum_k g_k \right) \right| \leq \theta d \|f\|_2 \|g\|_2. \quad (10)$$

Proof. Alon gives a deterministic polynomial-time construction, on exactly every sufficiently large number M of vertices, of a d -regular graph with every nontrivial eigenvalue at most $2\sqrt{d-1} + o_d(1)$ in absolute value [3]. First choose the constant d large enough and then the finite threshold for M so that this is at most θd . Its bipartite double cover has the same adjacency matrix between its two vertex classes, hence the required singular-value bound. Decomposing f and g into their constant and mean-zero parts proves (10). $\square$

5 Proof of Theorem 1

Fix a small $\eta > 0$, to be chosen from δ , and take the frame in Lemma 2. Set

$$a = (1 - \eta)\rho, \quad b = (1 + \eta)\rho.$$

Choose $0 < \theta < a^2$, and let G be the graph in Lemma 3. For every edge $e = (j, k)$ define the rank-one measurement matrix

$$A_e = b_j b_k^*. \quad (11)$$

Let $X = uv^*$ have rank at most one. (The zero case is immediate.) With $f_j = |b_j^* u|$ and $g_k = |b_k^* v|$, the Frobenius pairing factors as

$$|\langle A_e, X \rangle_F| = |b_j^* u| |b_k^* v| = f_j g_k. \quad (12)$$

Tightness gives $\|f\|_2 = \|u\|_2$ and $\|g\|_2 = \|v\|_2$, while flatness gives

$$a\sqrt{M}\|u\|_2 \leq \|f\|_1 \leq b\sqrt{M}\|u\|_2, \quad a\sqrt{M}\|v\|_2 \leq \|g\|_1 \leq b\sqrt{M}\|v\|_2.$$

Apply (10) to the nonnegative vectors f, g . Since $\|uv^*\|_F = \|u\|_2 \|v\|_2$, it follows that

$$d(a^2 - \theta)\|X\|_F \leq \sum_{e \in E} |\langle A_e, X \rangle_F| \leq d(b^2 + \theta)\|X\|_F. \quad (13)$$

Put $\alpha = d(a^2 - \theta)$ and $\beta = d(b^2 + \theta)$. Multiplying every A_e by $2/(\alpha + \beta)$ turns (13) into (3) with

$$\delta' = \frac{\beta - \alpha}{\beta + \alpha} = \frac{b^2 - a^2 + 2\theta}{a^2 + b^2}. \quad (14)$$

First choose η and then θ sufficiently small that $\delta' \leq \delta$. The number of edges is $m = dM$, and d depends only on δ . Lemma 2 gives (4). Every matrix in (11) is rank one and all ingredients are deterministic and polynomial-time constructible. $\square$

6 Why the proof does not yet extend beyond rank one

Six focused upgrade routes were checked after obtaining the rank-one theorem. They delimit the current result rather than form part of its proof.

Attempt 1. SVD plus triangle inequalities. For $X = \sum_{s=1}^r \sigma_s u_s v_s^*$, the upper estimate follows with $\|X\|_*$ and hence loses at most $\sqrt{r}$. The lower estimate fails because the summands cancel inside each scalar measurement before the absolute value is taken.

Attempt 2. Dense tensor coordinates first. Keeping all M^2 entries of BXB^* yields a deterministic embedding at the trivial quadratic scale. One can prove a constant lower bound by applying the flat estimate to columns and averaging scalar projections. This does not reduce the measurement count below $n^{2+o(1)}$.

Attempt 3. Ordinary spectral sampling of $|BXB^*|$. For rank one, this nonnegative matrix is an outer product. For rank $r > 1$, entrywise absolute value can have full rank. The generic trace-norm estimate in the mixing error loses $\sqrt{M}$; compensating for it forces degree of order M and returns to M^2 measurements.

Attempt 4. Positive semidefinite restriction. For $X \succeq 0$, tight-frame quadratic measurements sum to $\text{tr } X$. This only gives $\|X\|_F \leq \text{tr } X \leq \sqrt{r}\|X\|_F$, not a prescribed small RIP constant, and it does not treat indefinite or nonsymmetric matrices.

Attempt 5. Phase lifting and sign-rank samplers. The identity $|z| = \sup_{|\omega|=1} \text{Re}(\omega z)$ introduces an independent phase at every sampled edge. Those edgewise phases destroy the rank- r factorization. A sampler uniform over this phase-enlarged class would be a genuinely stronger pseudorandom object than a spectral expander.

Attempt 6. Polynomial/moment approximation to the absolute value. Approximating $|z|$ by polynomials in $|z|^2$ turns the problem into sampling higher tensor powers. Without a uniform entry bound, the approximation is not relative near zero; after truncation, the known tensor designs have degree or size exponential in the approximation degree or in r . No bound of the requested form $n^{\lambda} r^{\mu}$ emerges.

Thus the surviving obstruction is precise: a graph expander samples separable bilinear arrays, whereas $|BXB^*|$ is no longer separable after the rank-one case. A low-rank absolute-value sampler with polynomial dependence on r would upgrade the theorem, but no such lemma has been established here.

7 Verification, novelty bounds, and review recommendation

The accompanying script `code/verify_rank_one_expander.py` builds a finite exact flat frame from all sign vectors, a deterministic union-of-permutations bipartite graph, and the rank-one measurements (11). It checks tightness, the second singular value, the factorization identity, the raw constants in (13), and the rescaled RIP bounds on a fixed reproducible suite of real and complex samples. This is a regression check, not a proof; the proof is the two-lemma argument above.

The four lightweight run indexes were searched for arXiv:2310.18565, the title, “modified RIP,” “rank-one measurements,” deterministic low-rank recovery, tensor embeddings, and expander mixing. They contained adjacent records about deterministic sparse-vector RIP (arXiv:1003.2990, 1202.1234, and 1406.4089) but no statement of the theorem above. A bounded primary-source search of

arXiv and citation/phrase searches through 11 August 2026 located the source, Indyk’s embedding, standard explicit expanders, and randomized rank-one matrix RIP work, but no later paper stating this expander-sparsified rank-one construction. This is not an exhaustive novelty certification.

The main human-review focus should be Lemma 2: verify that the tight normalization is retained through the recursive block construction and terminal fixed-dimensional quadrature. The expander step is then a direct one-line application of mixing. The claimed status should remain *candidate substantial partial result* until that structural extraction and the bounded novelty search receive independent expert review.

References

- [1] S. Foucart, *Linearly Embedding Sparse Vectors from ℓ_2 to ℓ_1 via Deterministic Dimension-Reducing Maps*, arXiv:2310.18565 (2023), Section 6.
- [2] P. Indyk, *Uncertainty Principles, Extractors, and Explicit Embeddings of L_2 into L_1* , Electronic Colloquium on Computational Complexity, Report TR06-126 (2006); also STOC 2007. See Theorem 1.1 and Corollary 5.2.
- [3] N. Alon, *Explicit Expanders of Every Degree and Size*, Combinatorica 41 (2021), 447–463; arXiv:2003.11673.